# Linear Models Lecture 12: Hypothesis Testing

Robert Gulotty

University of Chicago

April 29, 2026

# Where We Are

**Last lecture:** We derived the Wald statistic

$$W = (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r})' \left[\boldsymbol{R}\hat{\boldsymbol{V}}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r}) \xrightarrow{d} \chi^2_q$$

and showed how to construct robust tests and confidence intervals.

**Today:** We go deeper into the *practice* of hypothesis testing.

- Constrained least squares: equality *and* inequality constraints
- Partial identification: when the data only pins $\boldsymbol{\beta}$ to a feasible set
- The F test (recap), Score (LM) tests, and the trinity of classical tests
- Test inversion for confidence regions
- Multiple testing and Bonferroni corrections
- Power: what determines whether you can detect an effect?
- Hansen's practical advice for applied work

# Constrained Least Squares: Setup

Many tests today compare an **unrestricted** model to a **restricted** one. We need to know how to estimate under restrictions.

**Problem:** Minimize the sum of squared errors subject to $q$ linear constraints:

$$\tilde{\beta}_{\text{CLS}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (Y_i - \boldsymbol{X}_i'\beta)^2 \quad \text{subject to} \quad \boldsymbol{R}'\beta = \boldsymbol{r}$$

where $\boldsymbol{R}$ is $k \times q$ and $\boldsymbol{r}$ is $q \times 1$.

**Examples:**

- $\beta_3 = 0$: set $\boldsymbol{R}' = (0, 0, 1, 0, \ldots)$, $\boldsymbol{r} = 0$ (exclusion restriction)
- $\beta_2 = \beta_3$: set $\boldsymbol{R}' = (0, 1, -1, 0, \ldots)$, $\boldsymbol{r} = 0$ (equality restriction)
- $\beta_2 + \beta_3 = 1$: set $\boldsymbol{R}' = (0, 1, 1, 0, \ldots)$, $\boldsymbol{r} = 1$ (adding-up constraint)

# The CLS Estimator

Solve via Lagrange multipliers. The solution is:

$$\boxed{\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}} - (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}\left[\boldsymbol{R}'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}\right]^{-1}(\boldsymbol{R}'\hat{\beta}_{\text{OLS}} - \boldsymbol{r})}$$

**Reading the formula:**

- Start from unrestricted OLS $\hat{\beta}_{\text{OLS}}$, subtract a correction toward the constraint
- Correction is proportional to $(\boldsymbol{R}'\hat{\beta}_{\text{OLS}} - \boldsymbol{r})$: how far OLS violates $H_0$
- If OLS already satisfies the restriction, $\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}$

The restricted residuals and variance estimate are:

$$\tilde{e}_i = Y_i - \boldsymbol{X}_i'\tilde{\beta}_{\text{CLS}}, \qquad \tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\tilde{e}_i^2$$

Since the restriction constrains the parameter space: $\tilde{\sigma}^2 \geq \hat{\sigma}^2$ always.

# Variance of the CLS Estimator (Robust)

For SEs and CIs on $\tilde{\beta}_{\mathrm{cls}}$ itself, we need its asymptotic variance. Hansen §8.7, no homoskedasticity assumed.

**Robust "meat" with restricted residuals.** Let $\hat{\boldsymbol{Q}}_{XX} = \frac{1}{n}\boldsymbol{X}'\boldsymbol{X}$.

$$\tilde{\Omega} = \frac{1}{n-k+q}\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i' \, \tilde{e}_i^2, \qquad \tilde{\boldsymbol{V}}_\beta = \hat{\boldsymbol{Q}}_{XX}^{-1} \, \tilde{\Omega} \, \hat{\boldsymbol{Q}}_{XX}^{-1}.$$

The Lecture 11 sandwich, but with $\tilde{e}_i$ and an $n-k+q$ d.f. adjustment.

**Project onto the constraint.** With $\tilde{\boldsymbol{A}} = \hat{\boldsymbol{Q}}_{XX}^{-1}\boldsymbol{R}\left(\boldsymbol{R}'\hat{\boldsymbol{Q}}_{XX}^{-1}\boldsymbol{R}\right)^{-1}\boldsymbol{R}'$ — the matrix that maps $\hat{\beta}_{\mathrm{ols}}$ to $\tilde{\beta}_{\mathrm{cls}}$:

$$\boxed{\tilde{\boldsymbol{V}}_{\mathrm{cls}} = (\boldsymbol{I} - \tilde{\boldsymbol{A}}) \, \tilde{\boldsymbol{V}}_\beta \, (\boldsymbol{I} - \tilde{\boldsymbol{A}})'}$$

Hansen §8.7 unfolds this into four terms. SE for $\boldsymbol{h}'\tilde{\beta}_{\mathrm{cls}}$ ($\boldsymbol{h}$ outside the row space of $\boldsymbol{R}$): $s(\boldsymbol{h}'\tilde{\beta}_{\mathrm{cls}}) = \sqrt{n^{-1}\boldsymbol{h}'\tilde{\boldsymbol{V}}_{\mathrm{cls}}\boldsymbol{h}}$.

# CLS in Practice

**Simple case:** If $H_0$ sets some coefficients to zero, CLS just means dropping those regressors.

```
# Unrestricted
mod_U <- lm(lwage ~ education + exper + I(exper^2) +
               female + union, data = wages)
# Restricted (H0: beta_female = beta_union = 0)
mod_R <- lm(lwage ~ education + exper + I(exper^2),
               data = wages)
```

**General case:** For arbitrary linear restrictions (e.g., $\beta_2 = \beta_3$), reparameterize or use the closed-form formula.

The key outputs from CLS are $\text{SSE}_R = \sum \tilde{e}_i^2$ and $\tilde{\sigma}^2$, which we use to build the F and Score statistics.

# Worked Example: CLS for a Linear Restriction

**Setup.** Wage regression with separate returns to general experience and firm-specific tenure:

$$\log w_i = \beta_0 + \beta_1 \, \mathrm{educ}_i + \beta_2 \, \mathrm{exper}_i + \beta_3 \, \mathrm{tenure}_i + e_i.$$

Becker–Mincer human-capital theory predicts these have *different* returns: firm-specific human capital should pay more at the current employer than general experience does. We want to *test* this — formally, $H_0 : \beta_2 = \beta_3$ vs. $H_1 : \beta_2 \neq \beta_3$.

Important framing. We do not impose the restriction because we believe it. We compute the restricted estimator to obtain $\mathrm{SSE}_R$, which feeds the $F$ statistic that *tests* the restriction.

# Worked Example: Two Ways to Compute CLS

**Approach 1: Reparameterize.** Under $H_0$ the model becomes

$$\log w_i = \beta_0 + \beta_1 \, \text{educ}_i + \beta_2 \, (\text{exper}_i + \text{tenure}_i) + e_i,$$

which is just OLS on the new combined regressor:

```
mod_R <- lm(lwage ~ educ + I(exper + tenure), data = wages)
```

**Approach 2: Apply the boxed CLS formula directly** with $\boldsymbol{R}' = (0, 0, 1, -1)$ and $\boldsymbol{r} = 0$.

Both give the same $\tilde{\beta}_{\text{CLS}}$ and the same $\text{SSE}_R = \sum \tilde{e}_i^2$, which feed the $F$ statistic developed later in this lecture.

# Beyond Equality: Inequality Constraints

The CLS formulation extends naturally to inequalities:

$$\tilde{\beta}_{\text{CLS}} = \arg\min_{\boldsymbol{\beta} \in \mathcal{B}} \sum_{i=1}^{n} (Y_i - \boldsymbol{X}_i' \beta)^2$$

where the **feasible set** $\mathcal{B} \subseteq \mathbb{R}^k$ is defined by linear inequalities $\boldsymbol{R}'\beta \leq \boldsymbol{r}$.

**Examples:**

- **Sign restrictions**: $\beta_j \geq 0$ (a variance, a price-elasticity sign, a share)
- **Box constraints**: $0 \leq \beta_j \leq 1$ (mixing weights, probabilities)
- **Monotonicity**: $\beta_1 \leq \beta_2 \leq \beta_3$ (ordered effects)
- **Theory bounds**: $\boldsymbol{\beta} \in [\underline{\boldsymbol{\beta}}, \overline{\boldsymbol{\beta}}]$ from prior knowledge

The feasible set $\mathcal{B}$ is a *polyhedron* (an intersection of half-spaces) rather than a single point or a linear subspace.

# Solving Inequality CLS: Two Cases

**Strategy.** Solve unconstrained OLS first, then check whether the result satisfies the constraints.

**Case 1: $\hat{\beta}_{\mathbf{OLS}} \in \mathcal{B}$.** No constraint binds; the constrained and unconstrained problems have the same solution:

$$\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}.$$

**Case 2: $\hat{\beta}_{\mathbf{OLS}} \notin \mathcal{B}$.** The constrained solution sits on the boundary of $\mathcal{B}$. On the face where one or more constraints bind, those constraints hold with equality, and the solution is given by the *equality*-CLS formula from earlier in this lecture, applied to the binding constraints only.

# Inequality CLS as a Projection

Conceptually, you are projecting $\hat{\beta}_{\text{OLS}}$ onto the feasible region $\mathcal{B}$.

Picking which constraints bind is what makes the inequality case harder than the equality case — in practice this is solved by a *quadratic program*. In R: `quadprog::solve.QP()`.

# Why Inequality Constraints? Partial Identification

Sometimes the data and minimal assumptions only tell us $\beta$ lies in a *set*, not a point.

**The identified set:** $\Theta_I = \{\beta : \beta \text{ is consistent with the data and assumptions}\}$.

- **Point identification**: $\Theta_I = \{\beta_0\}$ — a single value
- **Partial identification**: $\Theta_I$ is a non-degenerate set, often a polyhedron

**Where it shows up:** missing data, censoring, interval-valued outcomes, treatment effect bounds, set-identified moment inequalities.

The identified set *is* a feasible set $\mathcal{B}$. Estimation under partial identification reports both a point (e.g., the inequality-CLS solution on $\mathcal{B}$) and the *width* of $\mathcal{B}$ — and the width does *not* shrink with $n$.

# A Worked Example: NSW Job Training

The **National Supported Work Demonstration** (1976–77) randomly assigned disadvantaged workers to a 9–12 month training program ($X_i = 1$) or a control group ($X_i = 0$). LaLonde (1986) used it to benchmark non-experimental evaluators.

**Outcome.** $Y_i = 1$ if subject $i$ had positive earnings in 1978 (employed at follow-up).

**Sample** (Dehejia–Wahba experimental sample, $n = 445$):

| | $n$ | Employed in 1978 | $\bar{Y}_{\text{obs}}$ |
|---|---|---|---|
| Treatment ($X = 1$) | 185 | 140 | 0.757 |
| Control ($X = 0$) | 260 | 168 | 0.646 |

# NSW: Naive ATE and a Missing-Data Thought Experiment

**Naive ATE:** $\bar{Y}_{\text{obs}}(1) - \bar{Y}_{\text{obs}}(0) = 0.757 - 0.646 = 0.111$ (an 11 pp employment effect).

NSW's actual follow-up was nearly complete. **What if 30% had been lost?** The next slides treat that as a thought experiment to see what the data could still pin down.

# The Manski Decomposition (30% Missing)

Suppose only 70% of subjects could be located in 1978. Let $D_i = 1$ if observed; $P(D_i = 1) = 0.7$.

For each arm, by the law of total probability:

$$E[Y \mid X = x] \;=\; \underbrace{p(x)\,\bar{Y}_{\text{obs}}(x)}_{\text{identified}} + \underbrace{(1 - p(x))\,E[Y \mid X = x,\; D = 0]}_{\text{unknown}}$$

The first term is identified from observed respondents. The second is unknown — but since $Y \in \{0, 1\}$, the unknown conditional mean lies in $[0, 1]$, so:

$$\boxed{p(x)\,\bar{Y}_{\text{obs}}(x) \;\leq\; E[Y \mid X = x] \;\leq\; p(x)\,\bar{Y}_{\text{obs}}(x) + (1 - p(x))}$$

The lower bound replaces every missing $Y$ with 0 (worst case for that arm); the upper bound replaces every missing $Y$ with 1. These are **sharp**: no narrower bound is consistent with the data alone.

# Worst-Case Bounds: Apply to NSW

Plug in $p(x) = 0.7$ for each arm.

**Treatment** ($\bar{Y}_{\text{obs}} = 0.757$):

$$0.7 \cdot 0.757 \;\leq\; E[Y \mid X{=}1] \;\leq\; 0.7 \cdot 0.757 + 0.3 \quad\Longrightarrow\quad E[Y \mid X{=}1] \in [0.530,\, 0.830]$$

**Control** ($\bar{Y}_{\text{obs}} = 0.646$):

$$0.7 \cdot 0.646 \;\leq\; E[Y \mid X{=}0] \;\leq\; 0.7 \cdot 0.646 + 0.3 \quad\Longrightarrow\quad E[Y \mid X{=}0] \in [0.452,\, 0.752]$$

**ATE bounds.** $\theta = E[Y \mid X{=}1] - E[Y \mid X{=}0]$ is smallest when arm 1 is at its lower bound *and* arm 0 at its upper bound (largest in the opposite corner):

$$\theta \;\in\; [\,0.530 - 0.752,\;\; 0.830 - 0.452\,] \;=\; [-0.22,\;\; 0.38]$$

The naive estimate 0.11 sits inside, but the bound is wide enough that we cannot even sign the effect. Width = 0.60.

# Tighter Bounds via Monotone Selection

**Plausible assumption.** Subjects lost to follow-up — transient, moving, out of the labor force — are weakly *less* likely to be employed than those we observe in the same arm:

$$E[Y \mid X = x,\, D = 0] \;\le\; E[Y \mid X = x,\, D = 1] \;=\; \bar{Y}_{\text{obs}}(x).$$

**Effect.** The upper bound now replaces missing $Y$ with $\bar{Y}_{\text{obs}}(x)$, not with 1, so $E[Y \mid X{=}x] \le \bar{Y}_{\text{obs}}(x)$. The lower bound is unchanged.

**Apply to NSW:**

$$E[Y \mid X{=}1] \in [0.530,\, 0.757], \qquad E[Y \mid X{=}0] \in [0.452,\, 0.646],$$

$$\theta \;\in\; [\,0.530 - 0.646,\;\; 0.757 - 0.452\,] \;=\; [-0.12,\;\; 0.30].$$

Width drops from 0.60 to 0.42. Still crosses zero — an 11 pp underlying effect is too small for one mild assumption to settle the sign at 30% missing.

# The Identified Set as an Inequality-CLS Feasible Set

Write the CEF as a linear regression $E[Y \mid X] = \beta_0 + \beta_1 X$, so $\beta_0 = E[Y \mid X{=}0]$ and $\beta_0 + \beta_1 = E[Y \mid X{=}1]$.

The four MTS bounds map directly to four linear inequalities on $\boldsymbol{\beta} = (\beta_0, \beta_1)'$:

$$\mathcal{B} \;=\; \left\{\boldsymbol{\beta} : \;\; 0.452 \le \beta_0 \le 0.646, \;\; 0.530 \le \beta_0 + \beta_1 \le 0.757\right\}.$$

**Inequality CLS** on the observed sample:

$$\tilde{\boldsymbol{\beta}}_{\mathrm{CLS}} \;=\; \arg\min_{\boldsymbol{\beta} \in \mathcal{B}} \sum_{i:\, D_i = 1} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

$\mathcal{B}$ *is* the identified set: it pins $\beta_1$ to the interval $[-0.12, 0.30]$. An inequality-CLS point estimate is just one $\boldsymbol{\beta}$ inside $\mathcal{B}$; reporting that point hides the bound. Report the interval.

# Computing the Bounds in R

```
u  <- "https://users.nber.org/~rdehejia/data/nswre74_"
cn <- c("treat","age","educ","black","hispan","married",
        "nodegree","re74","re75","re78")
d  <- rbind(read.table(paste0(u,"treated.txt"), col.names=cn),
            read.table(paste0(u,"control.txt"), col.names=cn))
y1 <- mean(d$re78[d$treat==1] > 0)    # 0.7568   (n = 185)
y0 <- mean(d$re78[d$treat==0] > 0)    # 0.6462   (n = 260)
p  <- 0.7                              # 70% located, 30% missing
ate_wc  <- c(p*y1 - (p*y0+1-p), (p*y1+1-p) - p*y0)  # [-0.22, 0.38]
ate_mts <- c(p*y1 - y0, y1 - p*y0)                   # [-0.12, 0.30]
```

**What to report:** “The NSW employment ATE lies in $[-0.12, 0.30]$ under monotone selection (naive estimate 0.11).”

# Setting Up the Linear Program

Reparameterize from $\boldsymbol{\beta}$ to the conditional means $\boldsymbol{\mu} = (\mu_0, \mu_1)'$ where $\mu_x = E[Y \mid X{=}x]$.
Then $\beta_0 = \mu_0$ and $\beta_1 = \mu_1 - \mu_0 = \boldsymbol{c}'\boldsymbol{\mu}$ with $\boldsymbol{c} = (-1, 1)'$.

Each MTS bound is a row $\boldsymbol{a}_j'\boldsymbol{\mu}\ \{\geq, \leq\}\ r_j$. Stack the four:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}}_{A} \begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix} \quad \underbrace{\begin{matrix} \geq \\ \leq \\ \geq \\ \leq \end{matrix}}_{\texttt{dir}} \ \underbrace{\begin{pmatrix} 0.452 \\ 0.646 \\ 0.530 \\ 0.757 \end{pmatrix}}_{\texttt{rhs}}.$$

Why $\boldsymbol{\mu}$ and not $\boldsymbol{\beta}$? `lpSolve` defaults variables to $\geq 0$; conditional means satisfy this automatically, but $\beta_1$ can be negative.

# Bounds via lpSolve

Two LPs — minimize and maximize $\boldsymbol{c}'\boldsymbol{\mu} = \mu_1 - \mu_0$ over the same feasible set:

```
library(lpSolve)
A   <- rbind(c(1,0), c(1,0), c(0,1), c(0,1))
dir <- c(">=", "<=", ">=", "<=")
rhs <- c(0.452, 0.646, 0.530, 0.757)
c(b1_lo = lp("min", c(-1,1), A, dir, rhs)$objval,
  b1_hi = lp("max", c(-1,1), A, dir, rhs)$objval)
# b1_lo  b1_hi:  -0.116  0.305
```

lpSolve confirms $\beta_1 \in [-0.12, 0.30]$. More parameters or constraints just adds rows to $A$, dir, and rhs.

# The F Test (Recap from Section)

Compare the unrestricted SSE to the restricted (CLS) SSE. Testing $H_0 : \boldsymbol{R}'\beta = \boldsymbol{r}$ with $q$ restrictions:

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/q}{\mathrm{SSE}_U/(n-k)} = \frac{W^0}{q}$$

- **Distribution**: under normal homoskedastic errors, $F \sim F_{q,\,n-k}$ exactly; asymptotically (any iid errors), $qF \xrightarrow{d} \chi^2_q$.
- **Heteroskedasticity**: the $F_{q,n-k}$ distribution is no longer valid — use the robust Wald test from Lecture 11.
- **In R**: `anova(mod_R, mod_U)`.

> The package-default "F-statistic that all slopes equal zero" is rarely a scientifically interesting hypothesis. Hansen: there is no reason to report it.

# Score Tests: The Idea

The Wald test starts from the **unrestricted** estimate and asks: is $\hat{\boldsymbol{\theta}}$ far from $\boldsymbol{\theta}_0$?

The **Score test** (also called **Lagrange Multiplier test**) starts from the **restricted** estimate and asks: does the objective function want to move away from the restriction?

- Compute the restricted estimator $\tilde{\boldsymbol{\beta}}$ (constrained least squares)
- Evaluate the *gradient* (score) of the objective function at $\tilde{\boldsymbol{\beta}}$
- If $H_0$ is true, the score should be near zero (we're near the optimum)
- If $H_0$ is false, the score should be large (the restriction is pulling us away)

# Score Statistic for Linear Restrictions

For $H_0: \boldsymbol{R}'\beta = \boldsymbol{c}$ in the normal regression model:

$$S = \frac{(\boldsymbol{R}'\hat{\beta} - \boldsymbol{c})'[\boldsymbol{R}'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}]^{-1}(\boldsymbol{R}'\hat{\beta} - \boldsymbol{c})}{\tilde{\sigma}^2}$$

- Identical to $W^0$ except $s^2$ replaced by $\tilde{\sigma}^2$ (restricted variance).
- Under $H_0$: $S \xrightarrow{d} \chi^2_q$

**Key result:** $S$ is a monotone transformation of $F$: $S = n\left(1 - \frac{1}{1+qF/(n-k)}\right)$, so the Score test and the F test always give the same accept/reject decision.

## When Are Score Tests Useful?

For linear regression with linear restrictions, the Score test offers nothing new over $F$ or Wald.

**But** in more complex settings, Score tests have a major advantage:

- They only require estimation under $H_0$ (the restricted model)
- The "unrestricted" model may not even be a simple OLS regression

**Example:** Testing for heteroskedasticity (Breusch–Pagan). The null is homoskedastic OLS—easy. The unrestricted model allows $\text{Var}(e_i \mid X_i)$ to vary with $X$, which requires modeling the variance function. The Score test only needs the OLS residuals.

# Score Tests: More Settings Where They Help

The same logic — estimate only under $H_0$, evaluate the score — applies whenever the null gives a clean model and the alternative is messy.

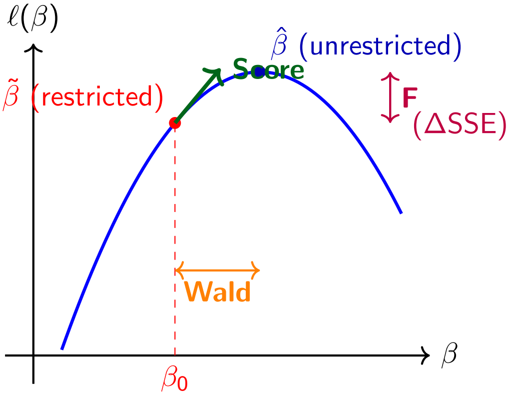
- **Serial correlation** (Breusch–Godfrey): null is OLS with iid errors.
- **Omitted nonlinearities**: null is a simple linear specification.
- Any setting where $H_0$ gives you a clean model but $H_1$ is messy.

# Three Ways to Test the Same Hypothesis

| | Wald | Score (LM) | F / LR-like |
|---|---|---|---|
| **Estimates from** | Unrestricted | Restricted | Both |
| **Measures** | Distance of $\hat{\boldsymbol{\theta}}$ from $\boldsymbol{\theta}_0$ | Gradient at restriction | Change in fit (SSE) |
| **Null dist.** | $\chi^2_q$ | $\chi^2_q$ | $\chi^2_q/q$ or $F_{q,n-k}$ |
| **Robust?** | Yes (sandwich) | Score-like | No |

For linear restrictions in the normal model, all three are monotone transformations of each other and give identical decisions. They differ for nonlinear restrictions or non-normal models.

# Visualizing the Trinity




- **Wald**: horizontal distance between $\hat{\beta}$ and $\beta_0$ **Score**: slope at restricted estimate **F**: vertical distance ($\Delta$SSE)

# Confidence Intervals *Are* Inverted Tests

Recall the standard 95% CI:

$$\hat{C} = \left[\hat{\theta} - 1.96 \cdot s(\hat{\theta}),\ \ \hat{\theta} + 1.96 \cdot s(\hat{\theta})\right]$$

This is exactly the set of values $\theta$ that are *not rejected* by a two-sided $t$-test:

$$\hat{C} = \{\theta \ : \ |T(\theta)| \leq 1.96\}$$

**General principle:** Inverting a test with good Type I error control produces a confidence set with good coverage.

$$P[\theta \in \hat{C}] = P[\text{Accept} \mid \theta] = 1 - P[\text{Type I error}]$$

# Why Test Inversion Matters: Nonlinear Parameters

Consider $\theta = \beta_1/\beta_2$ (e.g., peak experience in a wage equation).

**Approach 1:** Delta method CI

$$\hat{\theta} \pm 1.96 \cdot s(\hat{\theta})$$

From Lecture 11, this can be inaccurate for ratios (Fieller's problem).

**Approach 2:** Rewrite as a *linear* restriction $\beta_1 - \theta\beta_2 = 0$ and invert:

$$\hat{C} = \left\{ \theta \ : \ \frac{(\hat{\beta}_1 - \theta\hat{\beta}_2)^2}{\boldsymbol{R}'(\theta)\hat{\boldsymbol{V}}\boldsymbol{R}(\theta)} \leq 1.96^2 \right\}$$

where $\boldsymbol{R}(\theta) = (1, -\theta)'$. This requires a grid search over $\theta$.

# Test Inversion vs. Delta Method: Hansen's Example

**Setup.** Peak experience in a wage equation, $\theta = -50\beta_1/\beta_2$.

- **Delta method CI:** $[29.8,\ 29.9]$
- **Inverted linear-test CI:** $[29.1,\ 30.6]$

The delta method interval is **far too narrow** — the ratio's sampling distribution is poorly approximated by a normal centered at $\hat{\theta}$. Inverting the test for the equivalent linear restriction does not require that approximation.

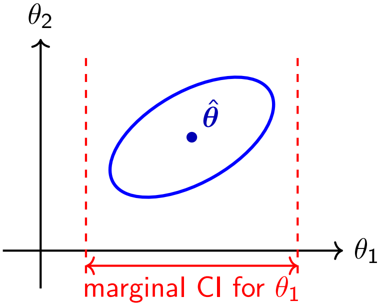
# Test Inversion in R

```
# Wage equation: log(wage) ~ exper + exper^2/100 + ...
mod <- lm(lwage ~ exper + I(exper^2/100) + education,
          data = wages)
b <- coef(mod); V <- vcovHC(mod)

# Grid search: invert the t-test for theta = -50*b1/b2
theta_grid <- seq(20, 40, by = 0.01)
in_CI <- sapply(theta_grid, function(th) {
  R <- c(0, 1, -th/50, 0)  # gradient of b[2] - th*b[3]/50
  num <- (b[2] - th * b[3] / 50)^2
  den <- t(R) %*% V %*% R
  num / den <= qchisq(0.95, 1)
})
CI <- range(theta_grid[in_CI])
```

# Confidence Regions for Multiple Parameters

For $q > 1$ parameters, invert the Wald test: $\hat{C} = \left\{\boldsymbol{\theta} : W(\boldsymbol{\theta}) \leq \chi^2_{q,\,1-\alpha}\right\}$ — an **ellipsoid** in $\mathbb{R}^q$ centered at $\hat{\boldsymbol{\theta}}$, shaped by $\hat{\boldsymbol{V}}_\theta^{-1}$.




Marginal CIs (projections) are always wider than the ellipsoid in each dimension.

# The Multiple Testing Problem

Suppose you test $k$ hypotheses, each at level $\alpha = 0.05$. Under the **global null** (all $k$ true), what is $P$[at least one false rejection]?

By **Boole's inequality**:

$$P\left[\min_{j \leq k} p_j < \alpha\right] \leq \sum_{j=1}^{k} P[p_j < \alpha] \rightarrow k\alpha$$

| $k$ (tests) | $\alpha$ per test | Familywise error $\leq$ |
|---|---|---|
| 5 | 0.05 | 0.25 |
| 10 | 0.05 | 0.50 |
| 20 | 0.05 | 1.00 |

With 20 tests, you are *virtually certain* to get a false rejection!

# The Bonferroni Correction

**Goal:** Control the *familywise error rate* (FWER) at $\alpha$.

**Rule:** Reject the $j$th hypothesis only if $p_j < \alpha/k$. Equivalently: $p_{\text{Bonf}} = k \cdot \min_{j \le k}\ p_j$.

**Proof:**

$$P\left[\min_{j \le k} p_j < \frac{\alpha}{k}\right] \le \sum_{j=1}^{k} P\left[p_j < \frac{\alpha}{k}\right] \to k \cdot \frac{\alpha}{k} = \alpha$$

**Simple and conservative.** Controls FWER for *any* dependence structure among the tests. The cost: reduced power when $k$ is large.

# Bonferroni: Worked Example

Two coefficients tested at $\alpha = 0.05$:

| | Individual $p$ | Bonferroni $p$ $(= 2 \times p)$ |
|---|---|---|
| Union membership | 0.04 | 0.08 |
| Married status | 0.15 | 0.30 |

- **Without correction:** Union membership "significant" at 5%.
- **With Bonferroni:** Neither significant at FWER $= 0.05$ (need $p < 0.025$).

**When to worry:** exploring many specifications/subgroups, examining many coefficients, or reporting the "most significant" result.

# Bonferroni in R

```
# Get p-values from a regression
mod <- lm(lwage ~ education + exper + I(exper^2) +
             female + union + married, data = wages)
pvals <- summary(mod)$coefficients[-1, 4]  # drop intercept

# Bonferroni correction
p_bonf <- p.adjust(pvals, method = "bonferroni")
cbind(raw = round(pvals, 4), bonferroni = round(p_bonf, 4))

# Other options: Holm (less conservative, still controls FWER)
p_holm <- p.adjust(pvals, method = "holm")
```

**Holm's method** is uniformly more powerful than Bonferroni while still controlling FWER. Use it when available.

# Type I and Type II Errors

| | $H_0$ **true** | $H_0$ **false** |
|---|---|---|
| **Reject** $H_0$ | Type I error ($\alpha$) | Correct (Power $= 1 - \beta$) |
| **Accept** $H_0$ | Correct ($1 - \alpha$) | Type II error ($\beta$) |

- **Size** ($\alpha$): probability of falsely rejecting a true null
- **Power** ($\pi$): probability of correctly rejecting a false null: $\pi(\theta) = P[\text{Reject } H_0 \mid \theta \neq \theta_0]$
- Power depends on the *true value* of $\theta$, on $n$, and on $\alpha$.

**Key trade-off:** Making $\alpha$ smaller reduces Type I errors but also reduces power. A test with $\alpha = 0$ never rejects and has zero power.

# Power of the $t$-Test

For $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$:

$$T = \frac{\hat{\theta} - \theta_0}{s(\hat{\theta})} \approx Z + \delta, \quad Z \sim N(0,1)$$

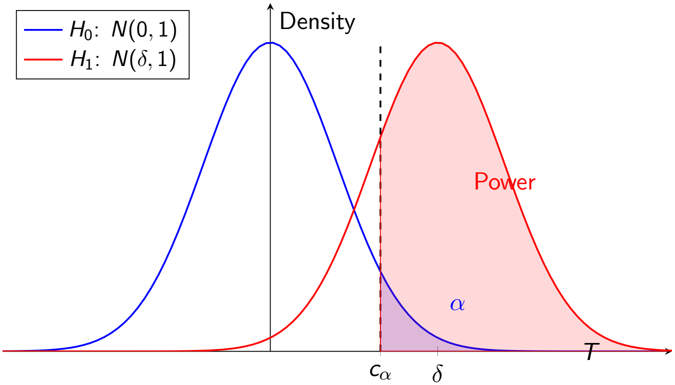
where $\delta = (\theta - \theta_0)/s(\hat{\theta})$ is the **signal-to-noise ratio**.

Power of a one-sided test at level $\alpha$: $\pi(\delta) = \Phi(\delta - z_\alpha)$

**Power increases when:**

- The true effect $|\theta - \theta_0|$ is larger
- The standard error $s(\hat{\theta})$ is smaller (more precise estimation)
- The significance level $\alpha$ is larger (less stringent threshold)
- The sample size $n$ is larger (since $s(\hat{\theta}) \propto 1/\sqrt{n}$)

# Visualizing Power



# Power in OLS: What You Can Control

In OLS, $s(\hat{\theta}) \approx \sigma_e/(\text{sd}(X) \cdot \sqrt{n})$ for a single regressor. So:

$$\delta \approx \frac{(\theta - \theta_0) \cdot \text{sd}(X) \cdot \sqrt{n}}{\sigma_e}$$

**Levers for increasing power:**

1. **Increase** $n$: power grows with $\sqrt{n}$
2. **Reduce** $\sigma_e$: add controls that explain $Y$ (reduce residual variance)
3. **Increase sd**$(X)$: more variation in the regressor of interest
4. **Increase** $\alpha$: use 5% instead of 1% (but more Type I errors)

> **Controls help power!** Relevant covariates reduce $\sigma_e$ without reducing sd$(X)$—a free power boost. But irrelevant covariates cost degrees of freedom.

# Power and Test Dimension

For joint tests ($q > 1$), the Wald statistic under local alternatives: $W \xrightarrow{d} \chi^2_q(\lambda)$, where $\lambda = \boldsymbol{h}'\boldsymbol{V}_\theta^{-1}\boldsymbol{h}$ is the **non-centrality parameter**.

**Critical fact:** For a fixed $\lambda$, power *decreases* as $q$ increases.

| $q$ (restrictions) | $\lambda$ for 50% power | Sample size increase |
|---|---|---|
| 1 | 3.85 | baseline |
| 2 | 4.96 | +28% |
| 3 | 5.77 | +50% |

**Takeaway:** Testing more restrictions simultaneously dilutes power. A single-coefficient $t$-test is more powerful than a joint $F$-test that includes other restrictions.

# The 50% Power Benchmark

How far must the true parameter be from $\theta_0$ for 50% power?

**One-sided test:**

- At $\alpha = 0.05$: need $\delta \geq 1.65$ standard errors
- At $\alpha = 0.01$: need $\delta \geq 2.33$ standard errors

**Implication for sample size:** $(2.33/1.65)^2 \approx 2$.

A test at $\alpha = 0.01$ requires roughly **twice the sample size** as $\alpha = 0.05$ to achieve the same power.

**Two-sided** ($\alpha = 0.05$): need $|\delta| \geq 1.96$ for 50% power. Since se $\propto 1/\sqrt{n}$, you need $n \propto (\sigma_e/(\theta - \theta_0))^2$.

# Power Calculation in R

```
# Approximate power for a two-sided t-test in OLS
# Inputs: effect size, residual SD, regressor SD, n, alpha
power_ols <- function(effect, sigma_e, sd_x, n, alpha=0.05) {
  se <- sigma_e / (sd_x * sqrt(n))
  delta <- effect / se
  z <- qnorm(1 - alpha/2)
  power <- pnorm(delta - z) + pnorm(-delta - z)
  return(power)
}

# Example: detect beta = 0.1 with sigma_e = 1, sd(x) = 2
sapply(c(50, 100, 200, 500, 1000), function(n)
  round(power_ols(0.1, 1, 2, n), 3))
# [1] 0.080 0.117 0.198 0.463 0.803
```

With $\beta = 0.1$, $\sigma_e = 1$, $\text{sd}(X) = 2$: you need $n \approx 1000$ for 80% power!

# Test Consistency

## Definition

*A test is* **consistent against fixed alternatives** *if for any true* $\theta \neq \theta_0$*:* $P[\text{Reject } H_0 \mid \theta] \to 1$ *as* $n \to \infty$.

**Good news:** The $t$-test and Wald test are consistent. As $n \to \infty$, $s(\hat{\theta}) \to 0$, so $|T| \to \infty$ whenever $\theta \neq \theta_0$.

**Caution:** Consistency means you will eventually reject any false null—including *economically trivial* deviations from $\theta_0$.

In very large samples, statistical significance $\neq$ economic significance. A statistically significant but tiny coefficient may not be meaningful.

# Hansen's Rules for Applied Work

1. **Report standard errors, not $t$-ratios.**
   - Standard errors focus attention on precision and confidence intervals.
2. **Report $p$-values, not asterisks.**
   - $p$-values contain more information than $*/{*}{*}/{*}{*}{*}$ ("an inferior practice").
3. **Focus on economically motivated hypotheses.**
   - Don't mechanically test every coefficient against zero.
   - Report the $t$-test for $\beta_j = 0$ when this is a scientifically interesting question.

# More Practical Advice

4. **"Do Not Reject" $\neq$ "Accept."**
   - Failing to reject means insufficient evidence, *not* that $H_0$ is true.
   - Never write: "the regression finds that $X$ has no effect on $Y$."
5. **Statistical significance $\neq$ economic significance.**
   - With large $n$, even tiny effects become "significant."
   - Always discuss the *magnitude* and substantive meaning.
6. **For nonlinear hypotheses, use minimum distance or test inversion.**
   - The Wald statistic is *not invariant* to the algebraic formulation of $H_0$.
   - Different equivalent formulations can give different results!

# The "What to Report" Checklist

For any coefficient of interest $\theta$:

1. **Point estimate** $\hat{\theta}$: the "best guess"
2. **Standard error** $s(\hat{\theta})$: measure of precision
3. **Confidence interval**: range of values consistent with the data
4. **Economic interpretation**: what the magnitude means in context

When testing:

5. State the **hypothesis** clearly (what question does this answer?)
6. Report the *p*-**value** (not just "significant" / "not significant")
7. Discuss **power** if you fail to reject (could you have detected a meaningful effect?)
8. Account for **multiple testing** if examining many hypotheses

# Summary

| Topic | Key Takeaway |
| --- | --- |
| CLS | Equality $\boldsymbol{R}'\beta = \boldsymbol{r}$ via Lagrange; gives $\text{SSE}_R$ for $F$ and Score |
| Inequality CLS | Feasible set $\mathcal{B}$ as a polyhedron; project OLS onto $\mathcal{B}$ via QP |
| Partial ID | $\Theta_I$ has width that does not vanish with $n$; report bounds, not just a point |
| F test | $F = W^0/q$; $\sim F_{q,n-k}$ under normal homo.; use robust Wald otherwise |
| Score test | From restricted model; equivalent to $F$ for linear restrictions |
| Trinity | Wald, Score, $F$ agree for linear $H_0$ in normal model |
| Test inversion | Confidence sets = non-rejected values; essential for nonlinear parameters |
| Bonferroni | With $k$ tests, use $\alpha/k$ to control familywise error |
| Power | $\delta =$ effect/se; grows with $\sqrt{n}$; 50% power needs $\delta \approx 2$ |
| Practical advice | Report SEs, CIs, $p$-values; focus on magnitudes; consider power |

# Next Time

**Lecture 13: Instrumental Variables**

- What happens when $E[\boldsymbol{Xe}] \neq \mathbf{0}$?
- Omitted variable bias, simultaneity, measurement error
- The IV solution: find $\boldsymbol{Z}$ such that $E[\boldsymbol{Ze}] = \mathbf{0}$ but $E[\boldsymbol{ZX}'] \neq \mathbf{0}$